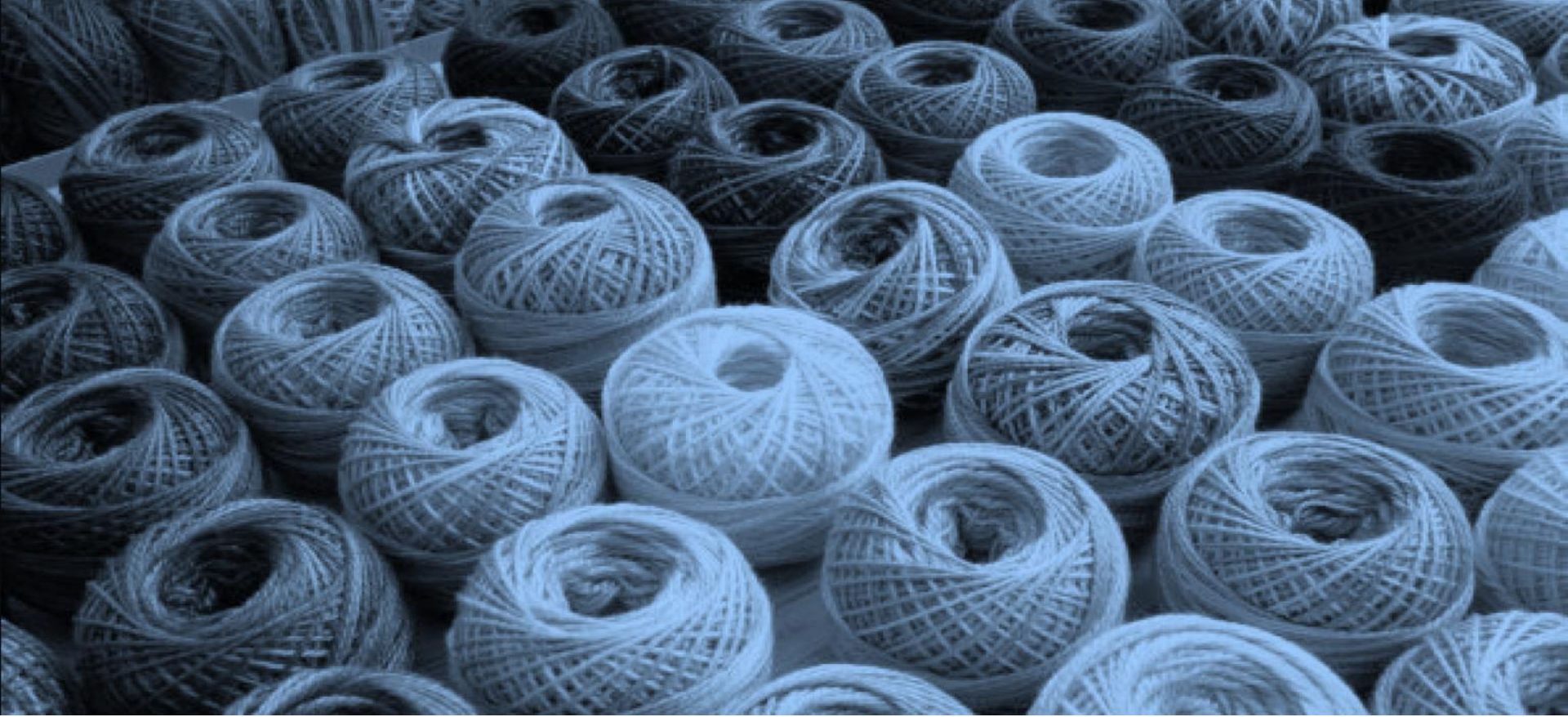


Java Programming

CPT111 – Lecture 5

Dr. Erick Purwanto

Xi'an Jiaotong-Liverpool University
西交利物浦大学

CPT111 Java Programming

Lecture 5

Static Methods 2 and String

Welcome to CPT111 – Lecture 5 !

- Find a friend to discuss today's lecture in classroom
- Please join: Lecture 5 In-Class Quiz
- Who is this character in **Chainsaw Man**?
  - I haven't read it
  - Denji
  - Power
  - Makima
  - I love Makima

Learning Outcomes

- Welcome to **Lecture 5** – Static Methods 2 and Strings
- **Learning Outcomes of this lecture**: we are going to learn about
  - More Static Methods
    - function definitions, implementations, calls
    - scope of variables
    - local variables vs global variables
  - Chars
    - char operations
  - Strings
    - primitive types vs reference types
    - string operations
    - string problem solving

Part 1: More Static Methods

- In Week 4 we have learned about static methods or functions
  - we will explore more about it this week

In-Class Quiz 1: Static Method from Last Week

- What does the static method **`myFunction`** do?

```java
public static int myFunction(int n) {
    int count = 0;
    for (int i = 1; i < n; i++) {
        if (i * i < n) {
            count++;
        }
    }
    return count;
}
```

  - counting odd numbers less than n
  - counting even numbers less than n
  - counting perfect squares less than n
  - counting perfect cubes less than n

Recall Week 4: Functions / Static Methods

- Example: Write a Java program that find the **maximum** of two integers
- You may write the following program in `main`:

```java
public class Lecture5Demo {
    public static void main(String[] args) {
        int x = 2, y = 5;
        if (x > y)
            System.out.println(x);
        else
            System.out.println(y);
        x = 10;  y = 8;
        if (x > y)
            System.out.println(x);
        else
            System.out.println(y);
    }
}
```

you test your code on the case where the second number is larger

then, you test your code on the case where the first number is larger

notice the repeated codes!

Recall Week 4: Static Methods

- We want to avoid **repeated** codes, and want to **organize** code as independent parts to share and **reuse** code to build bigger programs

```java
public class Lecture4Demo {

    public static int larger(int x, int y) {
        if (x > y) {
            return x;
        }
        return y;
    }

    public static void main(String[] args) {
        System.out.println(larger(2, 5));
        System.out.println(larger(10, 8));
    }
}
```

instead, write your program **once** inside a function/static method

now you can **call** your function in main **multiple** times!

Recall Week 4: Static Method Definition

- A static method consists of a **header** and a **body**
  - the body is enclosed in curly braces

```
public class Lecture4Demo {

    public static int larger(int x, int y) {
        if (x > y) {
            return x;
        }
        return y;
    }

    public static void main(String[] args) {
        System.out.println(larger(2, 5));
        System.out.println(larger(10, 8));
    }
}
```

give a descriptive **name** to a static method

specify the **parameter(s)** -> *input* to the function

specify the **type** of each *parameter* variable

specify the **type** of the *return / output* value

return the output value, do ***not*** println it -> **return statement(s)**

Recall Week 4: Static Method Call and Trace

- A static method/function call is method **name** followed by its **arguments**
  - separated by commas and enclosed in parenthesis

```java
public class Lecture4Demo {

    public static int larger(int x, int y) {
        if (x > y) {
            return x;
        }
        return y;
    }

    public static void main(String[] args) {
        System.out.println(larger(2, 5));
        System.out.println(larger(10, 8) + 5);
    }
}
```

function call to be printed

arguments must match the parameter type(s)

a function call is an *expression*, can be part of more complicated expressions

visualize using **Java Visualizer**!

Functions / Static Methods

- Java function / static method:
  - takes zero or more input *parameters*
  - *returns* zero or one output value
- Examples seen: `Math.random()`, `Integer.parseInt()`

0 input, 1 return value

1 input, 1 return value

- Example 1:

2 input, 1 return value

```
public static int larger(int x, int y) {
    if (x > y) {
        return x;
    }
    return y;
}
```

Functions / Static Methods

- Java function / static method:
  - takes zero or more input *parameters*
  - *returns* zero or one output value
- Example 2:

1 input, 0 return value

```
public static void printInts(int[] nums) {
    System.out.print("[");
    for (int i = 0; i < nums.length; i++) {
        if (i != nums.length-1)
            System.out.print(nums[i] + ", ");
        else
            System.out.print(nums[i]);
    }
    System.out.println("]");
}
```

Function Implementation

- To implement a function
  - create a *name*
  - declare type and name of *parameter(s)*
  - specify *type* for *return* value
  - implement *body* of method
  - finish with *return statement(s)*

```
public static int larger(int x, int y) {
    if (x > y) {
        return x;
    }
    return y;
}
```

function name

parameter declarations

return type

body of `larger()`

return statement(s)

Function Call

- To call a function
  - write the function name
    - if called from another class, write class name followed by `.` (e.g. `Math.random()`)
  - pass the *argument(s)* separated by commas in parentheses

```
System.out.println(larger(10, 8));
```

```
public static int larger(int x, int y) {
    if (x > y) {
        return x;
    }
    return y;
}
```

function name

arguments

Scope 作用域

- The *scope of a variable* is the part of the program that can refer to that variable by name
  - the scope of the variables declared in a block of statements is limited to the statements in that block
  - in particular, the scope of a variable declared in a static method is limited to that method's body
    - therefore, we cannot refer to a variable in one static method that is declared in another
    - we call this *local variable* 局部变量

Scope Example

- The *scope of a variable* is the part of the program that can refer to that variable by name

```
public static void printInts(int[] nums) {
    System.out.print("[");
    for (int i = 0; i < nums.length; i++) {
        if (i != nums.length-1)
            System.out.print(nums[i] + ", ");
        else
            System.out.print(nums[i]);
    }
    System.out.println("]");
}

public static void main(String[] args) {
    int[] arr = {1, 2, 3};
    printInts(arr);
}
```

scope of nums

scope of i

scope of args

scope of arr

In-Class Quiz 2 Method and Local Variables

- What is the output of the following Java program:

```
public static void negate(int a) {
    a = -a;
}

public static void main(String[] args) {
    int a = 5;
    System.out.println(a);
    negate(a);
    System.out.println(a);
}
```

- 5
  5
- 5
  -5
- -5
  5
- -5
  -5

Never Use Global Variables

- Best practice: declare variables so as to *limit their scope*

```
public class GlobalVariableDemo {

    public static int x;   // global variables
    public static int y;   // bad practice, don't do this

    public static int larger() {
        if (x > y) {
            return x;
        }
        return y;
    }

    public static void main(String[] args) {
        x = 2;
        y = 5;
        System.out.println(larger());
    }
}
```

Method Signature

- *Method signature* of a method consists of
  - method name
  - parameter types

method signature

```
public static int larger(int x, int y) {
    if (x > y) {
        return x;
    }
    return y;
}
```

Overloading

- Static methods whose signatures differ are different static methods
- Using the *same name* for two static methods whose *signatures differ* is known as *overloading*

```
public static int larger(int x, int y) {
    if (x > y)
        return x;
    return y;
}
```

can you implement `larger(int, int, int)` using `larger(int, int)`?

```
public static int larger(int x, int y, int z) {
    int max = x;
    if (y > max) max = y;
    if (z > max) max = z;

    return max;
}
```

Part 2: Char and String

- We continue our lecture today on `char` and `String`
  - and to solve problems using their supported operations

Primitive Data Type vs Reference Data Type

- A *data type* is a set of values and a set of operations defined on those values
- We have been learning the ***primitive*** *data types* that are built in Java
  - 8 primitive types: `byte`, `short`, `int`, `long`, `float`, `double`, `boolean`, **`char`**
- Everything else, including arrays and **`String`**, is *not* a primitive type but rather *a* ***reference*** *type*

  引用类型
  - we will see later that when we declare a reference type in a variable, we store not the data, but the reference / address / pointer of the data
- In Part 2 of today's lecture, we will learn more about `char` and consider `String` as a reference type, and also learn about string processing

Char

- A char is an alphanumeric character or symbol, enclosed in single quote ' '

```
char firstLetter = 'A';
char five = '5';
char newLine = '\n';
```

Char characters

- A char is a Unicode character
  - it is effectively an integer, ranges between 0 to 65,535 (inclusive)
  - it can be compared just like an integer, with `==` `!=` `<` `<=` `>=` `>`

```
char firstLetter = 'A';

firstLetter == 65;
```

Char Testing

- We can use `Character` wrapper class static methods to test a character:
  - `boolean isLetter(char c)` is c a letter?
  - `boolean isDigit(char c)` is c a digit?
  - `boolean isWhitespace(char c)` is c white space?
  - `boolean isUpperCase(char c)` is c an uppercase?
  - `boolean isLowerCase(char c)` is c a lowercase?

```
char firstLetter = 'A';
char five = '5';
char newLine = '\n';

Character.isLetter(firstLetter);
Character.isDigit(five);
Character.isWhitespace(newLine);
Character.isUpperCase(firstLetter);
Character.isLowerCase(firstLetter);
```

Uppercase, Lowercase, toString

- We can use `Character` class' static methods to change a character:
  - `char toUpperCase(char c)` to uppercase
  - `char toLowerCase(char c)` to lowercase
  - `String toString(char c)` to a `String` object consist of one character

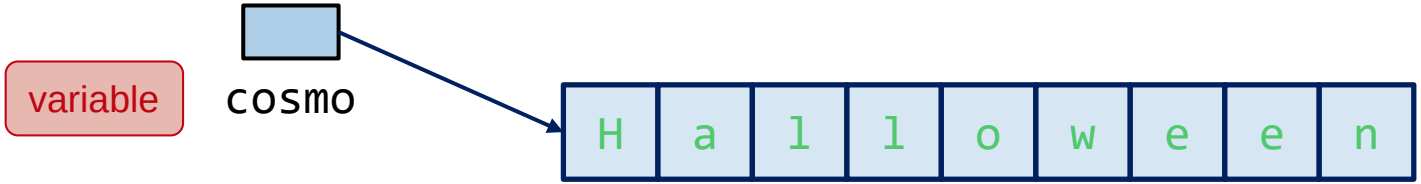
```
char firstLetter = 'A';
char smallA = 'a';

Character.toUpperCase(smallA);
Character.toLowerCase(firstLetter);
String oneLetterStr = Character.toString(firstLetter);
```

Java String

- A *Java string* is a sequence of characters (chars), like the word "Halloween", or a sentence "A corpse is talking."
- Create a Java string by writing its chars between double quotes " "

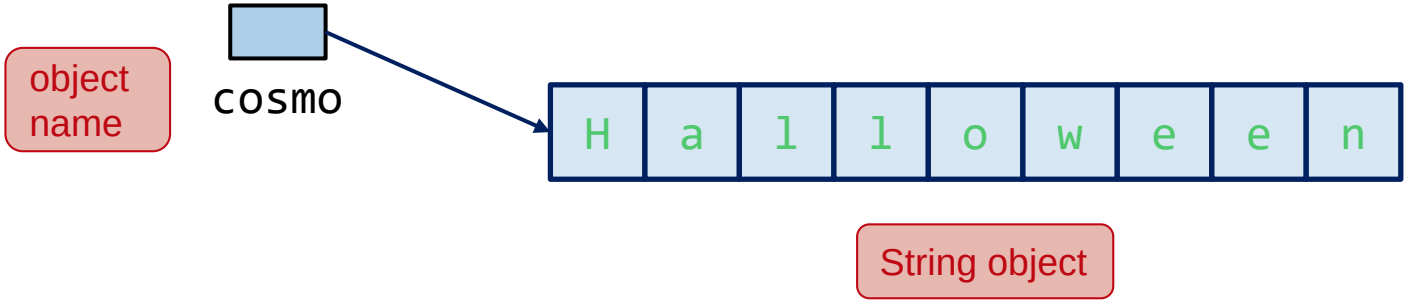
```
String cosmo = "Halloween";
```



String Class and Object

- String is a *class* or a *data type* in Java, its name is capitalized
- Create a String *object* by constructing a new *object*
  - use the keyword `new` to call a *constructor*
  - use *class* or *data type name* to specify type of object

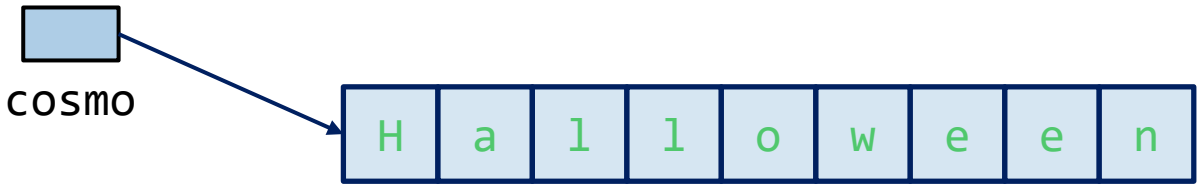
```
String cosmo;
cosmo = new String("Halloween");
```



Reference Variable Declaration

- When we declare a variable of any reference type such as String, Java allocates a box of 64 bits, no matter what type of object
  - the 64-bit box contains *not* the data about the string, but instead the *address* of the string in memory

```
String cosmo;
cosmo = new String("Halloween");
```



visualize using **Java Visualizer**!

Concatenation

- Plus + operator between strings concatenate them together to make a new, bigger string
  - chars of the first string put together with chars of the second string
  - works with strings stored in variables too

```
String cosmo = "hallo" + "ween";

String bang = "bang";
String exclamation = "!!";
String makima = bang + exclamation;
```

Length and Method Call

- The length of a string is the number of chars in it
- A *method* called `length()` on a string returns its length
- To call a method / to apply an *operation*:
  - use object name to specify which object
  - use the *dot operator* to indicate that an operation is to be applied
  - use a method name to specify which operation

```
String cosmo = "Halloween";
int len = cosmo.length();
System.out.println(len);
```

In-Class Quiz 3 Empty String and Length

- What is the output of the following Java program:

```
public static void main(String[] args) {
    String s = "";
    int len = s.length();
    System.out.println(len);
}
```

- ○ nothing is printed
- ○ 0 ✓
- ○ 1
- ○ 2
- ○ 3

Index Number

- The chars in a string are identified by *index numbers* `0`, `1`, `2`, ...
  - leftmost char is at index `0`
  - last char is at index `length-1`

H	a	l	l	o	w	e	e	n
0	1	2	3	4	5	6	7	8

Substring 1

- `substring` method picks out a part of string using index numbers to identify the desired part
- `substring(int start)` returns a new string made of the chars *starting at* index `start` and continue until the end of the string

H	a	l	l	o	w	e	e	n
0	1	2	3	4	5	6	7	8

```
String cosmo = "Halloween";
String ss1 = cosmo.substring(1);
String ss2 = cosmo.substring(2);
```

Substring 2

- Another version `substring(int start, int end)` returns a new string of the chars starting at index `start` up to but not including the end index
  - notice the length of the resulting substring can be computed by subtracting `end - start`

H	a	l	l	o	w	e	e	n
0	1	2	3	4	5	6	7	8

```
String cosmo = "Halloween";
String ss1 = cosmo.substring(2, 5);
String ss2 = cosmo.substring(8, 9);
```

llo

n

String Index Error (1)

- It is common to make mistakes with the index numbers fed into `substring()`
  - the valid index numbers for substring are `0`, `1`, `2`, ..., `str.length()`, so we need to be careful *not* to pass in numbers outside that range
    - the last number, `str.length()`, is one beyond the end of the string, used for the "***up to but not including***" index number end
- For example, suppose we want to take the first 5 chars of a string `str`
  - we may end up with String index out of bounds error for some `str`

```
String prefix = str.substring(0, 5);
```

String Index Error (2)

- To avoid out of bounds errors, add an if-statement to check the length of the string
  - don't assume that a string is long enough, check the `length()` before calling `substring()`

```
if (str.length() >= 5) {
    prefix = str.substring(0, 5);
}
else {
    // do something else when length is < 5
}
```

String Equality

- Use the `equals()` method to check if 2 strings are the same
  - `equals()` method is *case-sensitive* 区分大小写

same value

- Recall the == operator used to compare primitive types `int`, `double`, `char`
  - it *does not work reliably* with object types such as `String`

```
String cosmo = "Halloween";
String makima = "Bang";
if (cosmo.equals("Halloween")) {
    // correct use .equals() to compare Strings
}
if (cosmo == "Halloween") {
    // do not use == with Strings
}
cosmo.equals(makima);
```

String Equality Not Case Sensitive

- There is a variant of `String` equality check called `equalsIgnoreCase()` that compares two strings while ignoring uppercase/lowercase differences

```
String cosmo = "Halloween";

cosmo.equals("Halloween");
cosmo.equals("halloween");
cosmo.equalsIgnoreCase("halloween");
```

String Testing

- We can test with:
  - `boolean isEmpty()` is string an empty string?
  - `boolean contains(String substring)` does string contain `substring`?
  - `boolean startsWith(String prefix)` does string start with `prefix`?
  - `boolean endsWith(String postfix)` does string end with `postfix`?

```
boolean isEmpty() = cosmo.isEmpty();
```

```
String cosmo = "Halloween";

cosmo.isEmpty();
cosmo.contains("lowe");
cosmo.startsWith("hal");
cosmo.endsWith("ween");
```

Uppercase and Lowercase

- We can change string:
  - `String toUpperCase()` to uppercase
  - `String toLowerCase()` to lowercase

```
String cosmo = "Halloween";

cosmo.toUpperCase();
cosmo.toLowerCase();
```

For-Loop and String

- Use for-loop to iterate over the characters of a string
  - for example, loop to hit each index number once:

```
for (int i = 0; i < str.length(); i++) {
    // do something to str at index i

}
```

charAt

- `char charAt(int i)` returns the *character* at index `i`
  - for example, print each character of `str` once:

```
for (int i = 0; i < str.length(); i++) {
    // do something to str at index i
    System.out.println(str.charAt(i));
}
```

indexOf

- `str.indexOf(String target)` method searches left-to-right inside `str` for the string `target`
  - it returns the index number where the target string is *first found*, or `-1` if the target is *not found*
  - case-sensitive
  - you may use this instead of a for-loop to iterate over and look for a string

```
String cosmo = "Halloween";

int a = cosmo.indexOf("allo");
int b = cosmo.indexOf("e");
int c = cosmo.indexOf("allO");
```

lastIndexOf

- There is also `lastIndexOf(String target)` method that searches for the `target` from right to left

H	a	l	l	o	w	e	e	n
0	1	2	3	4	5	6	7	8

```
String cosmo = "Halloween";

int d = cosmo.lastIndexOf("e");
```

indexOf, lastIndexOf with fromIndex

- There are versions of `indexOf` and `lastIndexOf` that take a `fromIndex` to specify *where the search should begin*
  - `int indexOf(String target, int fromIndex)` : left-to-right, starts the search at the given `fromIndex`
  - `int lastIndexof(String target, int fromIndex)` : right-to-left, beginning at the given `fromIndex`
  - `fromIndex` does not actually need to be valid
    - if it is negative, the search happens from the start of the string
    - if it is greater than the string length, then -1 will be returned
- Using these, we can write a loop to find *all* the instances of the target in a string

Find all instances

- You may use `indexOf` to find all the instances of a target string, or just use the standard for-loop
- For example, we use `indexOf` in a loop to find all string `"Hallo"` in `str`
  - notice the `start` variable is used in the loop to keep resetting the starting point of the search

```
public static void findHallo(String str) {
    int start = 0;
    while (true) {
        int found = str.indexOf("Hallo", start);
        if (found != -1) {
            // found one, do something here
        }
        if (found == -1) break; // found none, need to stop looping
        start = found + 2;      // move start up for next iteration
    }
}
```

String Problem Solving Example 1

- Write a method to capitalize the text in parenthesis
  - given an input string with exactly one pair of parenthesis

```
public static String capitalParen(String str) {

    int left = str.indexOf("(");
    int right = str.indexOf(")");

    String sub = str.substring(left + 1, right);

    sub = sub.toUpperCase();

    String result = str.substring(0, left + 1) + sub +
            str.substring(right);

    return result;
}
```

Trim, Split, Replace

- Other popular string methods:
  - `String trim()` this string with leading and trailing whitespace removed
  - `String replace(String a, String b)` this string with a replaced by b
  - `String replaceAll(String a, String b)` this string with all a's replaced by b's
  - `String[] split(String delimiter)` array of strings between occurrences of `delimiter`

```
String cosmo = "  ha haa halloween  ";

cosmo = cosmo.trim();
cosmo = cosmo.replaceAll("h", "H");

String[] words = cosmo.split(" ");
for (int i = 0; i < words.length; i++) {
    System.out.println(words[i]);
}
```

String Problem Solving Example 2

- Write a method to validate a membership card id, with the following conditions:
  - Must be 13 characters long
  - Must be 4 blocks
  - Must be separated by hyphens
  - First character must be between A and D
    - can be upper or lower case
  - Other 3 blocks are 3 characters long
  - 2nd and 3rd blocks are all numbers
  - Final block must be 2 numbers and a letter
    - the letter must be between A and T

A-123-456-23A

A-187-267-111

c-542-223-11G

A-187-267+111

A-187-11G

E-123-456-23A

b-555-88-123T

String Validity Check

- Which one(s) do you think needs to be checked first?
  - ❑ Must be 13 characters long
  - ❑ Must be 4 blocks
  - ❑ Must be separated by hyphens
  - ❑ First character must be between A and D
  - ❑ Other 3 blocks are 3 characters long
  - ❑ 2nd and 3rd blocks are all numbers
  - ❑ Final block must be 2 numbers and a letter
  - ❑ The letter must be between A and T

A-123-456-23A

A-187-267-111

c-542-223-11G

A-187-267+111

A-187-11G

E-123-456-23A

b-555-88-123T

Skeleton Code and Testing Cases

- Start by writing an empty method (e.g. returning `true`), and write the test cases in main method

```
public static boolean checkMembership(String input) {
    // check conditions, any one unsatisfied, return false
    return true;
}

public static void main(String[] args) {
    String test1 = "A-123-456-23A";
    boolean valid = checkMembership(test1);
    System.out.println("Case 1 Expect: true Real: " + valid);
    String test2 = "A-187-267-111";
    valid = checkMembership(test2);
    System.out.println("Case 2 Expect: false Real: " + valid);
}
```

Checking Input Length

- Return `false` if any of the checking fails
  - Checking input length

```
public static boolean checkMembership(String input) {
    // check length
    if (input.length() != 13) {
        return false;
    }





    // passing all checks
    return true;
}
```

Checking Blocks and Delimiter

- Return `false` if any of the checking fails
  - Checking number of blocks and delimiter

```
public static boolean checkMembership(String input) {
    // check length
    if (input.length() != 13) {
         return false;
    }

    // check four blocks, separated by hyphens
    String[] blocks = input.split("-");
    if (blocks.length != 4) {
        return false;
    }

    // passing all checks
    return true;
}
```

In-Class Quiz 4: Have you learned something today?

- What does the static method myFunction do?

```
public static boolean myFunction(String s1, String s2, int n) {
    int count = 0;
    for (int i = 0; i < s1.length(); i++) {
        if (s1.char.At(i) != s2.charAt(i)) {
            count++;
        }
    }
    return count == n;
}
```

  - checking whether two strings have same length
  - checking whether two strings have different length
  - checking whether two strings differ by n chars at the same position
  - checking whether two strings differ by n chars at any position

Thank you for your attention !

- **Conclusion:** in this lecture, you have learned:
  - to create functions / static methods
  - about `char` primitive data type
  - about `String` reference data type
  - to use Character and String methods to do problem solving

- Please continue to **Lab 5**